Olivia Verge

Mechatronics Engineering Student — Biomedical Systems, Wearable Tech, & Embedded Software
613-698-4763 obverge@mun.ca linkedin.com/in/olivia-verge Portfolio

Education

Memorial University of Newfoundland **Sept. 2024 – Apr. 2029**
Bachelor of Engineering (B.Eng.), Mechatronics Engineering (Co-op) *St. John's, NL*
- Focus: Biomedical sensing, embedded software, human centered assistive design.'
- President, Memorial University Engineering Society 'B'
- Mechatronics Class Representative (Class of 2029) & Student Mentor

Experience

Nokia **Sept. 2026 – Present**
Mechanical Hardware Engineer (Co-op) *Ottawa, ON*
- Developing a software platform to automatically determine life cycle analysis of products, determining their material composition.
- Implementing cross platform API integrations to export data from individual CAD models and drawings
- Updating specialized assembly drawings for the production of new prototypes, and ensuring accurate change management documentation.

Memorial Medtech **May 2026 – Present**
Software & Electrical Technical Member *St. John's, NL*
- Contributed to early-stage software and electrical planning for sensor measurements and placement for PPG monitoring.
- Reviewed codebase of the teams award winning Atlantic ACL, winner of (True North Biomedical Competition) to build familiarity with its software architecture.
- Developed the team website for the James Dyson Award application, to showcase the Atlantic ACL

Mysa **Jan. 2026 – May 2026**
Hardware Product Engineer (Co-op) *St. John's, NL*
- Developed a Python automation pipeline leveraging Cursor AI to generate parametric STL and STEP models, reducing manual CAD modeling time by 30%.
- Managed product life cycle data using Arena PLM, executing 15+ ECOs and maintaining complex BOMs to support hardware releases.
- Supported new product introduction (NPI) through rapid SLA/FDM prototyping, design validation, and cross-functional engineering reviews.

Software & Technical Projects

ClearHead: Concussion Risk Monitor | *ROS2, Python, WebSockets* **Mar. 2026 – Apr. 2026**
- Developed a ROS2 publish/subscribe architecture to process 6-DOF IMU telemetry and detect high-risk impacts.
- Built a real-time JavaScript dashboard with a WebSocket bridge for live sensor telemetry and automated alerts.

PostureGuard: Biometric Computer Vision App | *Python, OpenCV, MediaPipe* **Feb. 2026 – Mar. 2026**
- Built a cross-platform computer vision application for real-time biomechanical monitoring using MediaPipe pose estimation, achieving accurate slouch detection.
- Implemented personalized baseline calibration and geometric landmark analysis for classification of spinal alignment.
- Integrated asynchronous notifications and system tray execution for macOS and Windows.

Adaptive Accessibility Controller | *CAD, 3D Printing, Electronics* **Jan. 2025 – Apr. 2025**
- Engineered a custom one-handed Nintendo Switch controller by rewiring electronics in a custom enclosure.
- Utilized human-centered design and iterative prototyping to deliver a functional solution for limited-mobility users.

Technical Skills

Programming: Python, C++, Javascript, React, SQL, Matlab

Software: Git, Github, Docker, ROS2, OpenCV, Mediapipe

Engineering Tools: Creo, Solidworks, Arena PLM, Aras Innovator, PSpice

Engineering: Product Development Testing, BOM Management, ECOs, NPI, Technical Documentation, Signal Processing

Hardware: Arduino, Sensors, IMUs, Circuit Analysis, Soldering, 3D Printing

Languages: English, French